

Harshit Kumar

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Summary

Backend engineer with 3 years building Go-based fintech infrastructure and distributed systems. Author of [Engram](#), an open-source cognitive memory system for AI agents. Deep interest in AI infrastructure, agent systems, and high-throughput backend architecture.

Skills

Languages: Go, JavaScript, TypeScript, Dart · **Backend:** Node.js, Nest.js, REST, WebSockets, gRPC

AI Infrastructure: pgvector, RAG, vector search, LLM application architecture, agent memory systems

Cloud & Infra: AWS (EC2, ECS, ALB, S3), Terraform, Kubernetes, Docker, Linux · **Databases:** PostgreSQL, MongoDB, Redis

Observability: Prometheus, Grafana, structured logging · **Other:** WebRTC, FFmpeg, RTMP

Work Experience

Senior Software Engineer · [Finplify](#)

Aug 2025 – Present

- Architected and built Go-based fintech backend handling high-value transactions end-to-end across booking, payment processing, and document registration workflows.
- Designed reconciliation and ledger systems ensuring accuracy, auditability, and consistency of user funds across 10+ daily transactions.
- Managed AWS infrastructure (EC2, ECS, S3, Secrets Manager), improving deployment reliability and reducing incident frequency by 30%.
- Built monitoring, logging, and alerting pipelines, cutting mean time to detection (MTTD) from hours of debugging to 1–2 minutes.
- Developed automation systems for notification, alerting, and operational workflows, reducing manual ops effort by ~80%.

Software Engineer · [Grappus](#)

Jul 2023 – Jul 2025

- Shipped 10+ production backend services in Go and Node.js, serving 50k+ users across multiple client systems.
- Optimized MongoDB queries and indexing strategies, reducing read latency by ~4x and infra costs by ~50%.
- Migrated infrastructure from EC2 to ECS with ALB, eliminating deployment downtime and cutting deploy time from 10 to 1 minute.
- Implemented observability stack (Prometheus + Grafana) for real-time monitoring and alerting across 5+ services.
- Built backend systems for AI-driven internal tools, including evaluation pipelines and automated workflow orchestration.

Projects

Engram · Cognitive infrastructure for AI agents

[engram.to](#) · [Deep dive](#)

- Open-source cognitive memory system for AI agents implementing belief dynamics (confidence, reinforcement, contradiction) and temporal decay — a fundamentally different approach than vector-only retrieval.
- Designed multi-layer memory retrieval combining vector search with structured recall, improving contextual reasoning across long-horizon agent interactions.
- Implemented learning loops enabling agents to adapt behavior based on outcomes and historical interactions.
- Built production-ready APIs, background workers, and scalable architecture in Go with PostgreSQL (pgvector).

DNS Server (Go)

[GitHub](#)

- Built a DNS server in Go supporting UDP and DNS-over-HTTPS (DoH), with caching and reverse DNS resolution.
- Developed CLI tooling for record queries with advanced debugging and verbosity controls.

Live Streaming Service

[GitHub](#) · [Write-up](#)

- Built low-latency streaming system using Go, WebRTC, and Next.js for real-time media delivery.
- Implemented RTMP ingestion and media pipelines with transcoding, optimizing performance and reducing server load.
- Designed real-time event updates and webhooks for stream lifecycle management.

Open Source Contributions

Active contributor to infrastructure and developer-tooling projects: [Glasskube](#) (Kubernetes package manager, 2.5k+ stars) · [Distr](#) (distributed application delivery) · [Hive](#)

Writing

- *[Memory is not a vector database: why AI agents need beliefs, not storage](#)* — on the limitations of vector-only memory and the case for belief-driven architectures.
- *[Integrating RTMP and WebRTC for real-time streaming](#)* — on building low-latency streaming infrastructure.

Education

B.Tech, Computer Science · Jaypee Institute of Information Technology, Noida

2023